SkinScan AI

System Analysis & Design · UI / UX Prototyping

Problem Statement · Use Cases · Requirements · Software Architecture · Wireframes · Design System · Accessibility

Project	AI-Powered Skin Condition Detection — Web Application
Track	AI & Data Science — Microsoft ML Engineer
Group Code	ONL4_AIS2_S7
Team	NHA-4-188
Instructor	Eman Elgalad
Date	May 2026

Why SkinScan AI exists

1.1 Problem Statement

Skin conditions affect over 1.9 billion people worldwide, making dermatology one of the most common areas of concern in primary healthcare. Despite this scale, access to qualified specialist care remains critically limited across many regions.

Challenge	Impact
Limited Specialist Access	In many parts of Africa and rural Asia, a single dermatologist serves over one million patients, making specialist consultations nearly impossible for most of the population.
Extended Wait Times	In numerous countries the average wait time to see a dermatologist exceeds 30–60 days, during which a condition may worsen considerably.
Late Detection Risk	Melanoma identified at an advanced stage carries a 5-year survival rate of only 27%, whereas early-stage identification yields a survival rate exceeding 98%.
Condition Misidentification	Patients frequently confuse visually similar conditions such as eczema, psoriasis, and ringworm, leading to self-treatment that aggravates the underlying issue.
Telemedicine Bottleneck	Existing telemedicine platforms require a licensed clinician to review every case individually, limiting scalability and response speed.

An intelligent, accessible screening tool that evaluates photographic evidence alongside patient-reported symptoms can address this gap — helping triage cases, prioritising urgent referrals, and equipping users with evidence-based self-care guidance, without replacing the role of a medical professional.

1.2 Project Objectives

ID	Objective	Description
O-01	Accessible Screening	Provide an instant first-pass screening tool enabling users to receive an informed assessment without requiring an immediate clinical appointment.
O-02	Multi-Condition Coverage	Support detection and differentiation of clinically relevant skin conditions such as Eczema, Warts Molluscum, Atopic Dermatitis, Benign Keratosis.
O-03	Multimodal Input	Accept photographic images, written symptom descriptions input simultaneously to maximize usability and assessment accuracy.
O-04	Triage & Referral	Automatically assign a triage level (Home Care, see a doctor, or URGENT) and generate condition-specific care guidance or referral instructions.
O-05	Visual Explainability	Accompany every assessment with a visual heatmap highlighting the regions that most influenced the result, supporting user trust.

ID	Objective	Description
O-06	Screening Assistance Only	Frame all outputs as screening assistance — never as a clinical assessment with mandatory disclaimers directing users to consult a qualified dermatologist.

02 · USE CASE DIAGRAM & DESCRIPTIONS

System actors and interactions

2.1 System Actors

Actor	Type	Role
Patient / End User	Primary Actor	Submits images and symptom information; receives assessment results and care guidance.
AI Screening Engine	System Actor	Processes inputs through the visual and language pipelines; generates assessment outputs.

2.2 Use Case Diagram

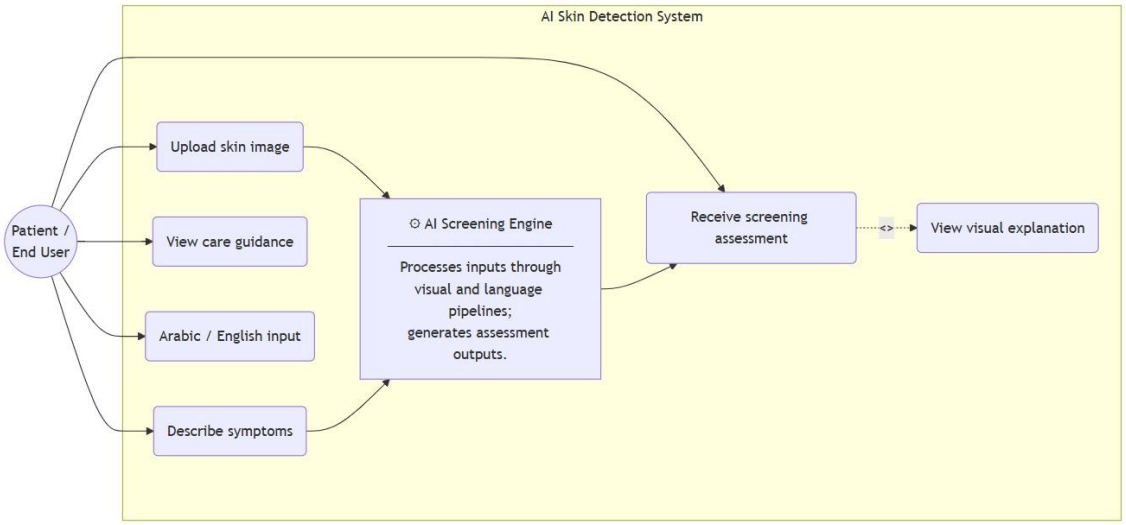


Figure 1 — Use case diagram showing actors, system boundary and supported interactions.

2.3 Use Case Descriptions

UC-01 — Upload Skin Image

Primary Actor(s)	Patient / End User
Description	Enables the user to submit a photograph of the affected skin area. The system provides real-time framing guidance to ensure image quality.
Precondition	User is logged in and has navigated to the screening interface.
Main Flow	1. User selects the option to upload an image or use the device camera. 2. The system displays an on-screen framing guide indicating optimal distance, lighting, and focus. 3. User captures or selects the image. 4. The system validates the image and confirms readiness for processing.
Alternative Flow	If image quality is insufficient, the system prompts the user to retake the photograph with specific guidance.
Postcondition	A validated image is queued and ready for processing.

UC-02 — Enter Symptom Description

Primary Actor(s)	Patient / End User
Description	Allows the user to describe their symptoms through typed text input. The system processes the description to extract relevant clinical indicators.
Precondition	An image has been uploaded.
Main Flow	1.For text: user types a description including duration, itch level, pain, spreading pattern, and known triggers. 2.The system displays the content for review and confirmation. 3.User confirms the symptom description to proceed.
Postcondition	A confirmed symptom description is ready for processing alongside the uploaded image.

UC-03 — Receive Screening Assessment

Primary Actor(s)	Patient / End User, AI Screening Engine
Description	The system processes the submitted image and symptom description through the AI Screening Engine and returns a ranked list of likely conditions with associated confidence levels.
Precondition	A validated image and confirmed symptom description have been submitted.

Main Flow	1. The AI Screening Engine processes the image through the visual analysis pipeline. 2. The engine simultaneously processes the symptom description through the language pipeline. 3. Both outputs are combined to generate a ranked list of likely conditions with confidence percentages. 4. The system presents the top result prominently, with secondary possibilities listed below. 5. If the user is not actively viewing the screen, the Notification Service delivers a summary alert.
Alternative Flow	If the system cannot produce a result with sufficient confidence, it returns an indeterminate result and recommends consulting a dermatologist.
Postcondition	The user has received a ranked screening assessment with condition names and confidence levels.

UC-04 — View Heatmap Explanation

Primary Actor(s)	Patient / End User
Description	Provides the user with a visual overlay on the submitted image highlighting the regions that most influenced the assessment result, supporting transparency and user trust.
Precondition	An assessment result has been generated and is displayed on screen.
Main Flow	1. User selects the option to view the visual explanation overlay. 2. The system renders a colour-coded heatmap overlay on the original submitted image. 3. Warmer colours indicate regions that most strongly influenced the assessment; cooler colours indicate minimal influence. 4. User can toggle the overlay on and off to compare the original image with the highlighted version.
Alternative Flow	Heatmap generation is always available for completed assessments.
Postcondition	User has reviewed the heatmap and understands which visible skin features contributed to the result.

UC-05— Receive Triage & Care Guidance

Primary Actor(s)	Patient / End User
Description	Based on the screening result and triage level assigned, the user receives actionable care guidance ranging from home care instructions to urgent referral messages.
Precondition	An assessment result with a triage level has been generated.

Main Flow	1. The system evaluates the result and assigns one of three triage levels: Home Care, See a Doctor, or URGENT. 2. Home Care: condition-specific self-care instructions are displayed. 3. See a Doctor: the system advises booking a dermatologist appointment within a specified timeframe. 4. URGENT: a prominent warning is displayed with immediate specialist referral instructions; the Notification Service sends an urgent alert. 5. All guidance screens include a mandatory disclaimer.
Postcondition	User has received and acknowledged condition-appropriate care guidance.

03 · FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

System capabilities and constraints

3.1 Functional Requirements

Image Submission

ID	Requirement
FR-05	The system shall accept photographic image uploads in JPEG and PNG formats.
FR-06	The system shall display real-time framing guidance for adequate lighting, focus, and distance.
FR-07	The system shall validate uploaded images by checking minimum resolution and confirming a detectable skin region before processing.

Symptom Input

ID	Requirement
FR-08	The system shall provide a free-text input field describing duration, intensity, known triggers, and spreading pattern.
FR-09	The system shall support voice input and integrate with a speech-to-text service to transcribe in real time.
FR-10	The system shall display the transcribed text for review and correction before submission.
FR-11	The system shall accept symptom input in multiple languages, including English and Arabic.

AI Screening

ID	Requirement
FR-12	The system shall combine visual and language pipeline outputs to produce a ranked list of likely conditions with confidence levels.
FR-13	The system shall present the top condition prominently with secondary possibilities listed below.

ID	Requirement
FR-14	The system shall return an indeterminate result with a referral recommendation when confidence is insufficient.

Triage & Care Guidance

ID	Requirement
FR-15	The system shall assign one of three triage levels: Home Care, See a Doctor, or URGENT.
FR-16	The system shall display condition-specific care instructions aligned with the assigned triage level.
FR-17	All assessment outputs shall include a mandatory disclaimer clarifying that the result is screening assistance only.
FR-18	URGENT triage levels shall trigger an alert notification to the user's registered email or device.

Visual Explanation

ID	Requirement
FR-19	The system shall generate a heatmap overlay on the submitted image highlighting influential regions.
FR-20	The user shall be able to toggle the heatmap on and off to compare against the original image.

3.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Return a completed assessment within three seconds of submission under standard broadband.
NFR-02	Privacy	Submitted images shall not be retained beyond the processing window unless the user opts in to storage.
NFR-03	Usability	The screening workflow shall be completable in five steps or fewer without prior training.
NFR-04	Accessibility	The user interface shall conform to WCAG 2.1 Level AA standards across all primary screens.

High-level system design

4.1 Architecture Style — Microservices

The system adopts a **Microservices Architecture**. Each functional concern — user management, image processing, language processing, result fusion, notification delivery, and administration — is implemented as an independent, separately deployable service. This style was chosen over a monolithic or MVC layout to enable independent scalability of the AI inference services, fault isolation between pipelines, technology flexibility per service, and maintainability across teams.

4.2 Architecture Diagram

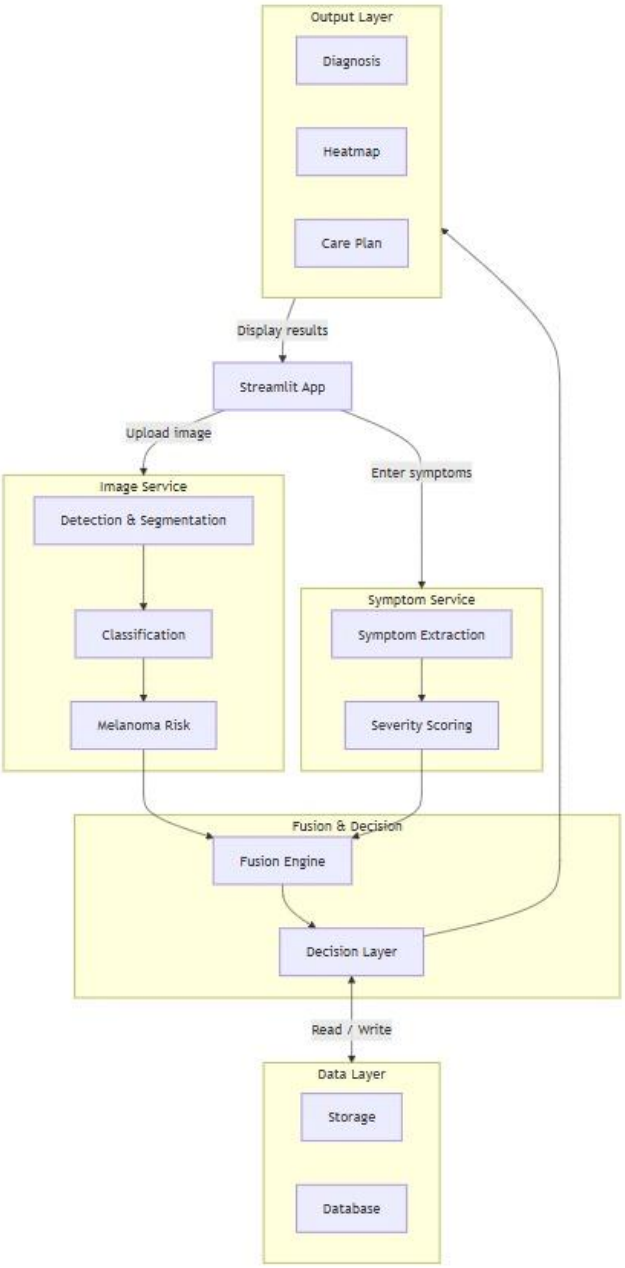


Figure 2 — High-level component architecture showing the client, API gateway, application services, vision and NLP pipelines, fusion / decision layer, output layer, and data layer.

4.3 System Components

Component	Layer	Description
Streamlit App	Presentation Layer	The front-end web interface through which the patient uploads skin images, enters symptom descriptions, and receives assessment results. Serves as the central orchestrator, routing inputs to the Image Service and Symptom Service, and displaying outputs from the Output Layer.
Image Service	Processing Layer	Handles all image-based analysis. Accepts uploaded images from the Streamlit App and executes a three-stage pipeline: Detection & Segmentation identifies the region of interest; Classification assigns condition labels; Melanoma Risk scoring evaluates malignancy probability.
Detection & Segmentation	Image Service	Isolates the affected skin region from the background to ensure downstream models focus on clinically relevant pixels, reducing noise and improving classification accuracy.
Classification	Image Service	Applies a convolutional neural network to the segmented region and maps visual features to one of six target conditions: Acne Vulgaris, Atopic Eczema, Psoriasis, Melanoma Risk, Ringworm, or Rosacea.
Melanoma Risk	Image Service	A specialised sub-model that evaluates asymmetry, border irregularity, colour variation, and diameter to produce a dedicated malignancy risk score, supplementing the primary classification output.
Symptom Service	Processing Layer	Processes the patient's textual symptom description through two sequential steps: Symptom Extraction identifies clinically relevant keywords and phrases; Severity Scoring quantifies urgency based on reported symptom characteristics such as duration, spread, and pain level.

Symptom Extraction	Symptom Service	Uses a natural language processing model to parse the symptom description and extract structured clinical indicators including symptom type, onset, location, and associated factors.
Severity Scoring	Symptom Service	Assigns a numerical severity score to the extracted symptom profile. The score informs the Fusion Engine's triage decision and is used to weight symptom data against image-based findings.
Fusion Engine	Fusion & Decision Layer	Combines the outputs of the Image Service and Symptom Service into a unified confidence vector. Applies a weighted aggregation strategy to balance visual and textual evidence before passing the result to the Decision Layer.
Decision Layer	Fusion & Decision Layer	Interprets the fused confidence vector to assign a final condition label, a confidence percentage, and a triage level (Home Care, See a Doctor, or URGENT). Passes the decision to both the Output Layer and the Data Layer.
Output Layer	Presentation Layer	Composes and delivers the three primary outputs returned to the user: Diagnosis (condition name and confidence score), Heatmap (visual explainability overlay on the submitted image), and Care Plan (triage-specific guidance and referral instructions).
Data Layer	Persistence Layer	Provides persistent storage for all assessment records, user profiles, and audit logs via a structured Database. The Fusion & Decision Layer reads and writes to this layer to support session continuity, dermatologist review, and administrative reporting.

4.4 End-to-End Data Flow

1. User Input (Streamlit App) The patient opens the web app and submits two inputs: an uploaded skin image and a typed symptom description.

2. Image Pipeline (Image Service) The image is routed to the Image Service where it goes through three stages — Detection & Segmentation isolates the skin region, Classification identifies the condition, and Melanoma Risk scoring evaluates malignancy probability.

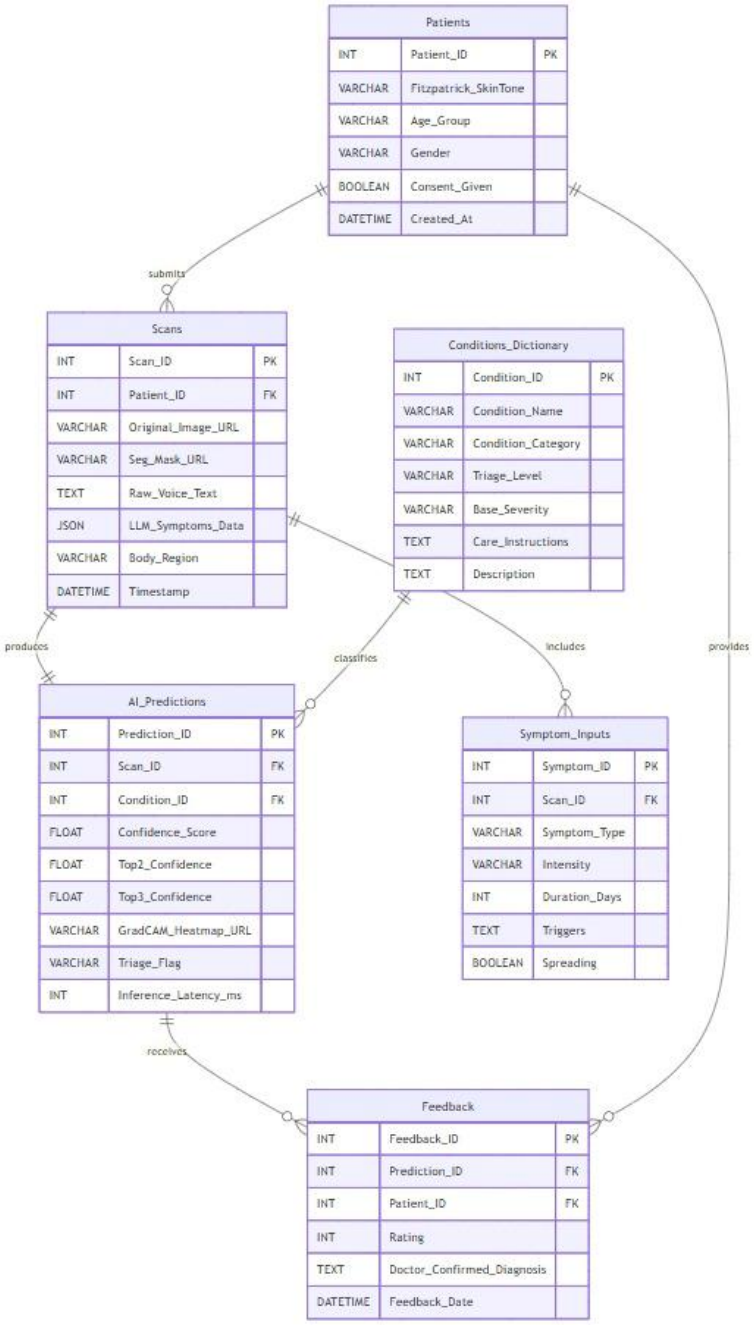
3. Symptom Pipeline (Symptom Service) The symptom text is simultaneously routed to the Symptom Service where Symptom Extraction pulls out clinical keywords, then Severity Scoring assigns a numerical urgency score.

4. Fusion (Fusion Engine) Both pipeline outputs — the image confidence scores and the symptom severity score — are merged by the Fusion Engine into a single unified confidence vector.

5. Decision (Decision Layer) The Decision Layer interprets that vector and produces three things: a final condition label, a confidence percentage, and a triage level (Home Care / See a Doctor / URGENT).

6. Output Delivery (Output Layer) The result is packaged into three outputs sent back to the Streamlit App: a Diagnosis, a Heatmap overlay on the original image, and a Care Plan with triage-specific guidance.

5.1 ER Diagram



Logical & Physical Schema

Overview The logical and physical schema is architected to support a high-performance, real-time Multimodal AI pipeline. It efficiently manages structured demographic data, unstructured medical inputs (audio/images), and the quantized models' outputs while maintaining strict adherence to clinical privacy and sub-3-second inference KPIs.

A. Tables, Attributes, and Keys

Table 1: Patients (Demographics & Compliance)

- Patient_ID (INT) - **[Primary Key]**: Unique identifier for the user.
- Fitzpatrick_SkinTone (VARCHAR): Captures the skin tone category (I-VI) to continuously monitor and mitigate algorithmic bias, ensuring skin tone fairness.
- Consent_Given (BOOLEAN): A mandatory flag ensuring explicit user consent is logged prior to any image processing or data retention.

Table 2: Scans (Multimodal Inputs)

- Scan_ID (INT) - **[Primary Key]**: Unique identifier for the consultation session.
- Patient_ID (INT) - **[Foreign Key]**: References Patients.Patient_ID.
- Original_Image_URL (VARCHAR): Cloud storage pointer for the raw uploaded image.
- Seg_Mask_URL (VARCHAR): Pointer to the isolated lesion mask generated in a single pass by the optimized **YOLOv8-Seg** model.
- Raw_Voice_Text (TEXT): The transcribed symptom description processed via the Whisper API.
- LLM_Symptoms_Data (JSON): A structured, schema-less payload storing parsed symptoms, triggers, and severity duration extracted simultaneously by the Lightweight Medical LLM.
- Timestamp (DATETIME): Automatically generated timestamp for the inference request.

Table 3: Conditions_Dictionary (Clinical Knowledge Base)

- Condition_ID (INT) - **[Primary Key]**: Unique identifier for the medical condition.
- Condition_Name (VARCHAR): The standardized clinical name (e.g., Acne Vulgaris, Atopic Eczema, Melanoma Risk).
- Base_Severity (VARCHAR): Categorical severity level mapping (Mild, Moderate, Severe, URGENT).
- Care_Instructions (TEXT): Condition-specific medical guidelines, OTC recommendations, and referral protocols.

Table 4: AI_Predictions (Model Outputs & Telemetry)

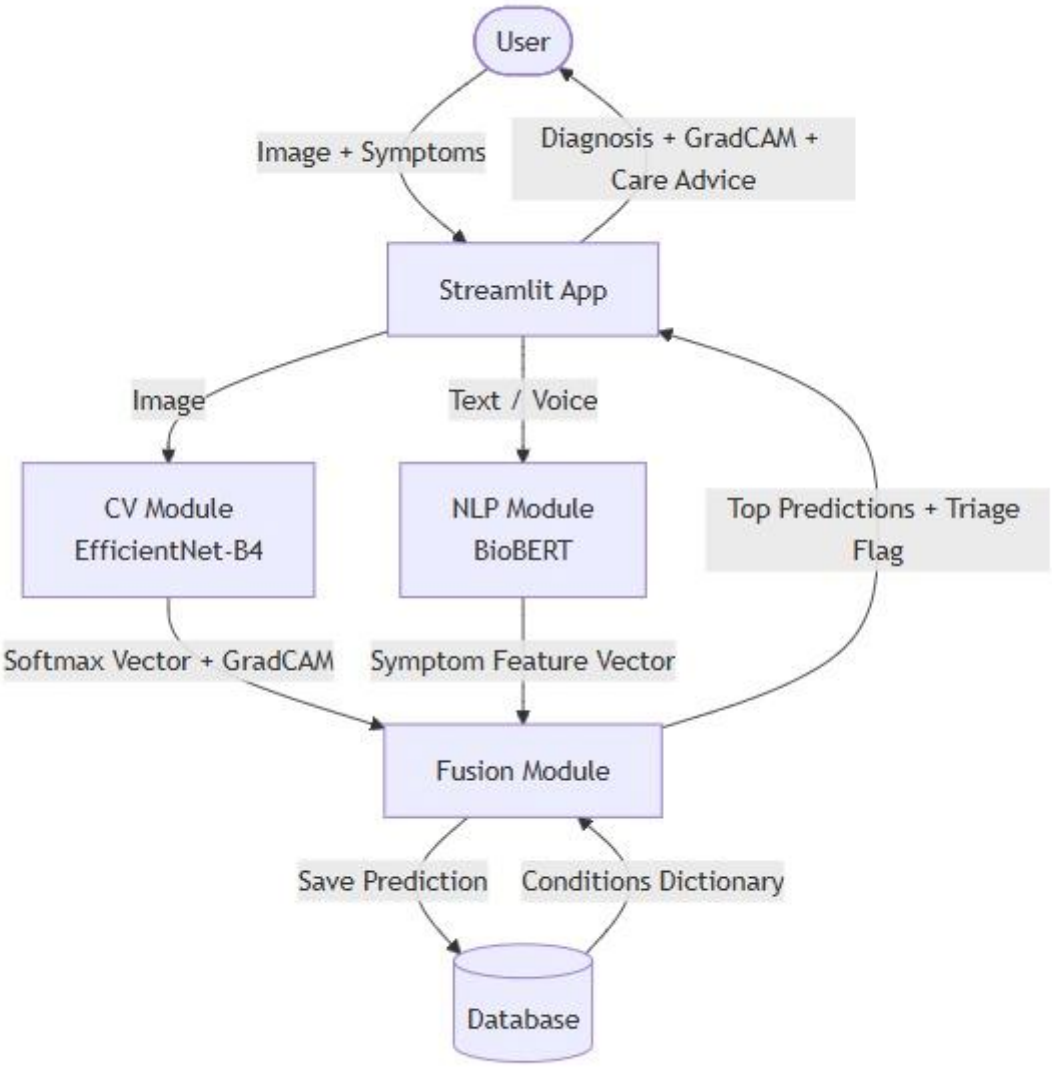
- Prediction_ID (INT) - **[Primary Key]**: Unique identifier for the diagnostic result.
- Scan_ID (INT) - **[Foreign Key]**: References Scans.Scan_ID.
- Condition_ID (INT) - **[Foreign Key]**: References Conditions_Dictionary.Condition_ID.
- Confidence_Score (FLOAT): The calibrated probability percentage output by the Multimodal Fusion module.
- ABCD_Melanoma_Risk (FLOAT): Dedicated continuous risk score generated by the parallel ResNet50 classifier.
- GradCAM_Heatmap_URL (VARCHAR): Storage pointer for the explainable AI visual overlay.
- Triage_Flag (VARCHAR): Automated routing decision (Home Care, See Doctor, URGENT).
- Inference_Latency_ms (INT): Tracks real-time processing speed to validate the success of the Model Quantization (ONNX/TensorRT) and ensure the sub-3-second KPI is met.

B. Normalization & Scalability Considerations

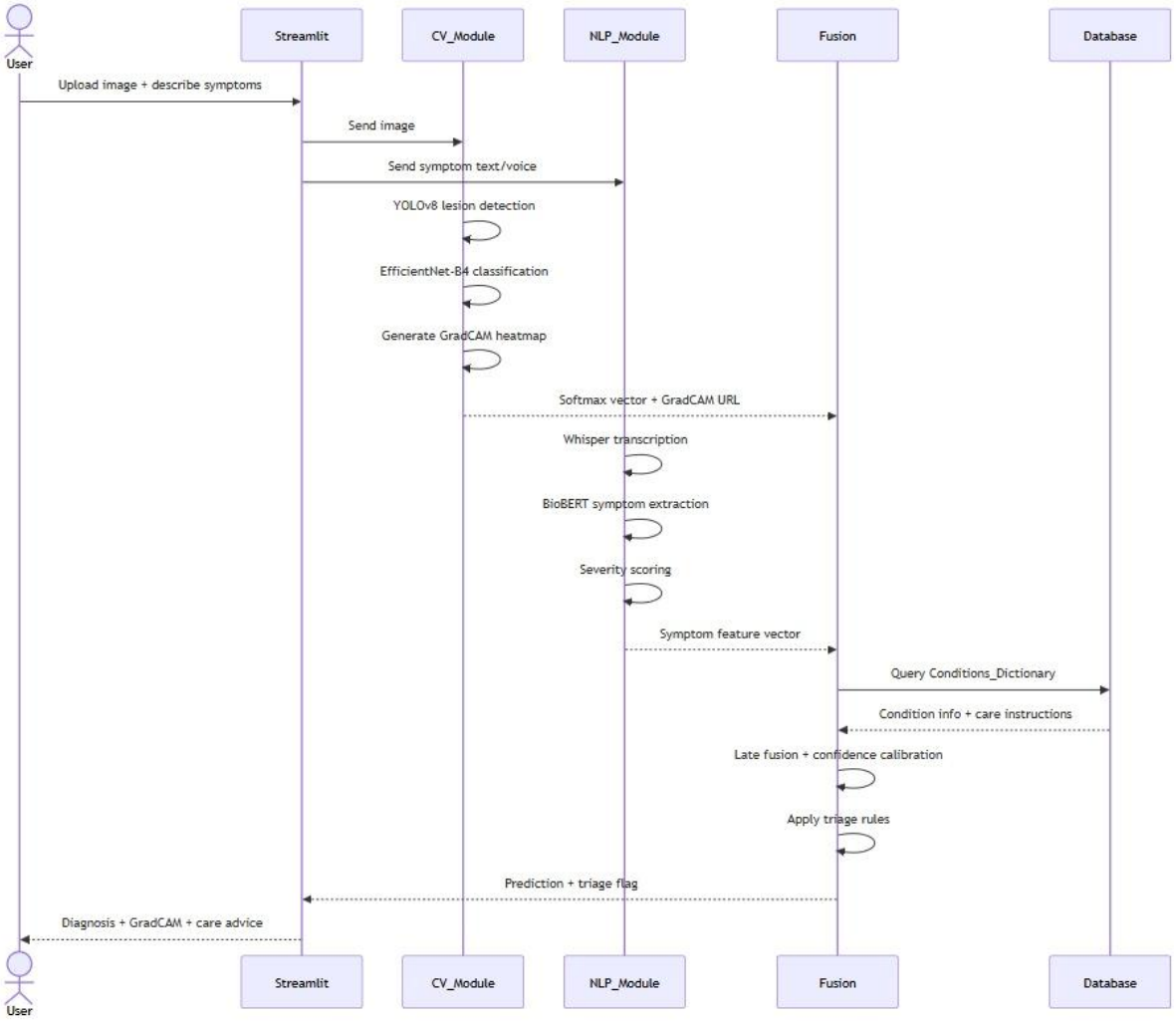
The database is strictly normalized up to the **Third Normal Form (3NF)** to eliminate redundancy and optimize the query execution plan during real-time inference:

1. **1NF/2NF & Heavy Asset Decoupling:** All relational attributes are atomic. To prevent database bloat and I/O bottlenecks, heavy binary files (Original Images, YOLO Masks, Grad-CAM Heatmaps) are entirely decoupled from the relational database. They are stored in an external object storage service (e.g., AWS S3), with only their lightweight VARCHAR URLs saved in the schema.
2. **3NF Dictionary Isolation:** Transitive dependencies were eliminated by creating the Conditions_Dictionary. Rather than duplicating static medical advice and triage rules for every single patient prediction, the AI_Predictions table utilizes a Foreign Key (Condition_ID). This drastically reduces storage costs and allows instantaneous updates to medical protocols across the entire system.
3. **JSON Schema-less Hybrid Integration:** Patient symptoms are highly variable (e.g., some report triggers, others report duration). By utilizing a JSON data type for LLM_Symptoms_Data in the Scans table, the database easily ingests the dynamic outputs of the Lightweight LLM without requiring dozens of sparse, NULL-filled columns, resulting in a highly scalable physical schema.

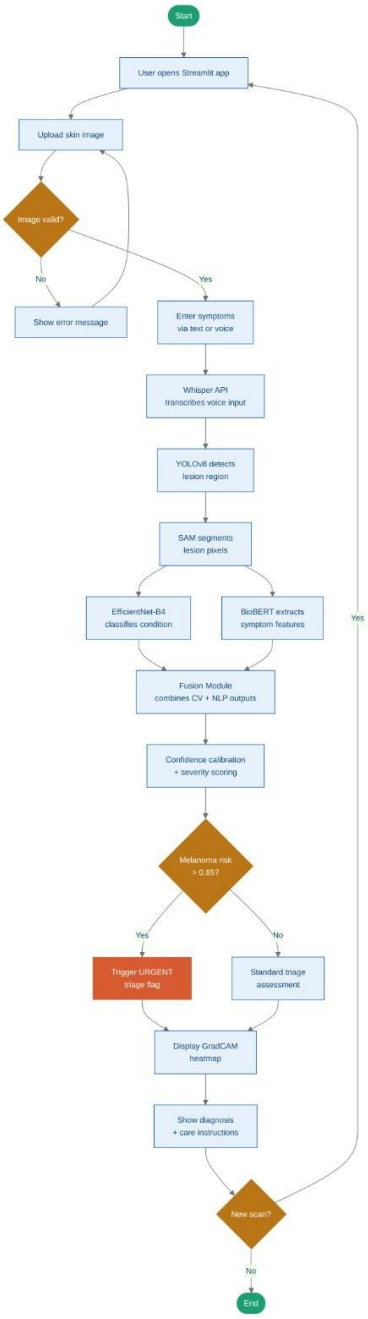
1. Data Flow Diagram



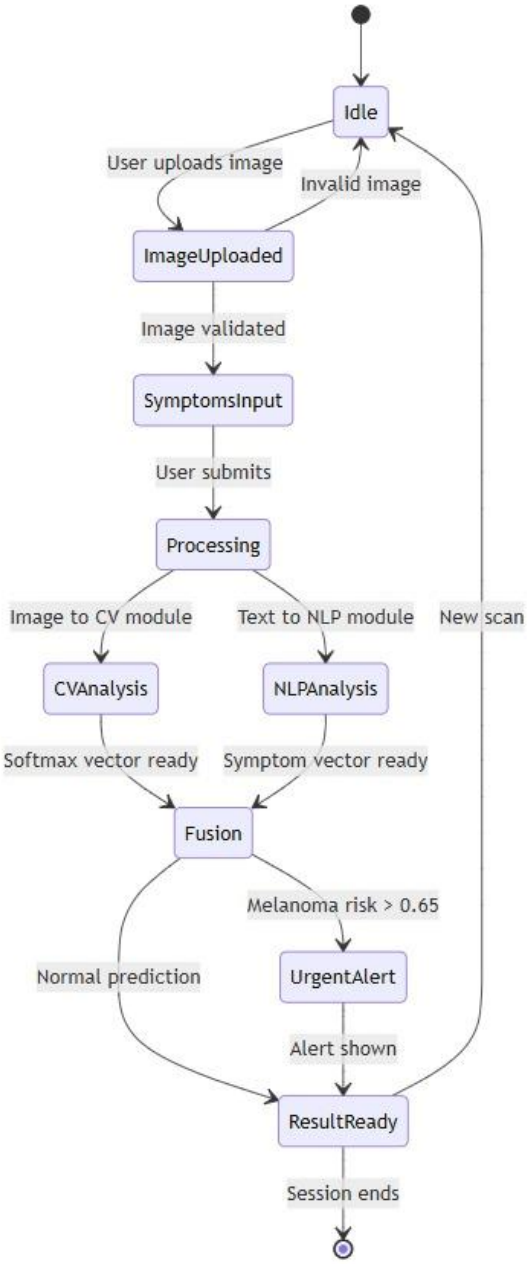
2.Sequence Diagram

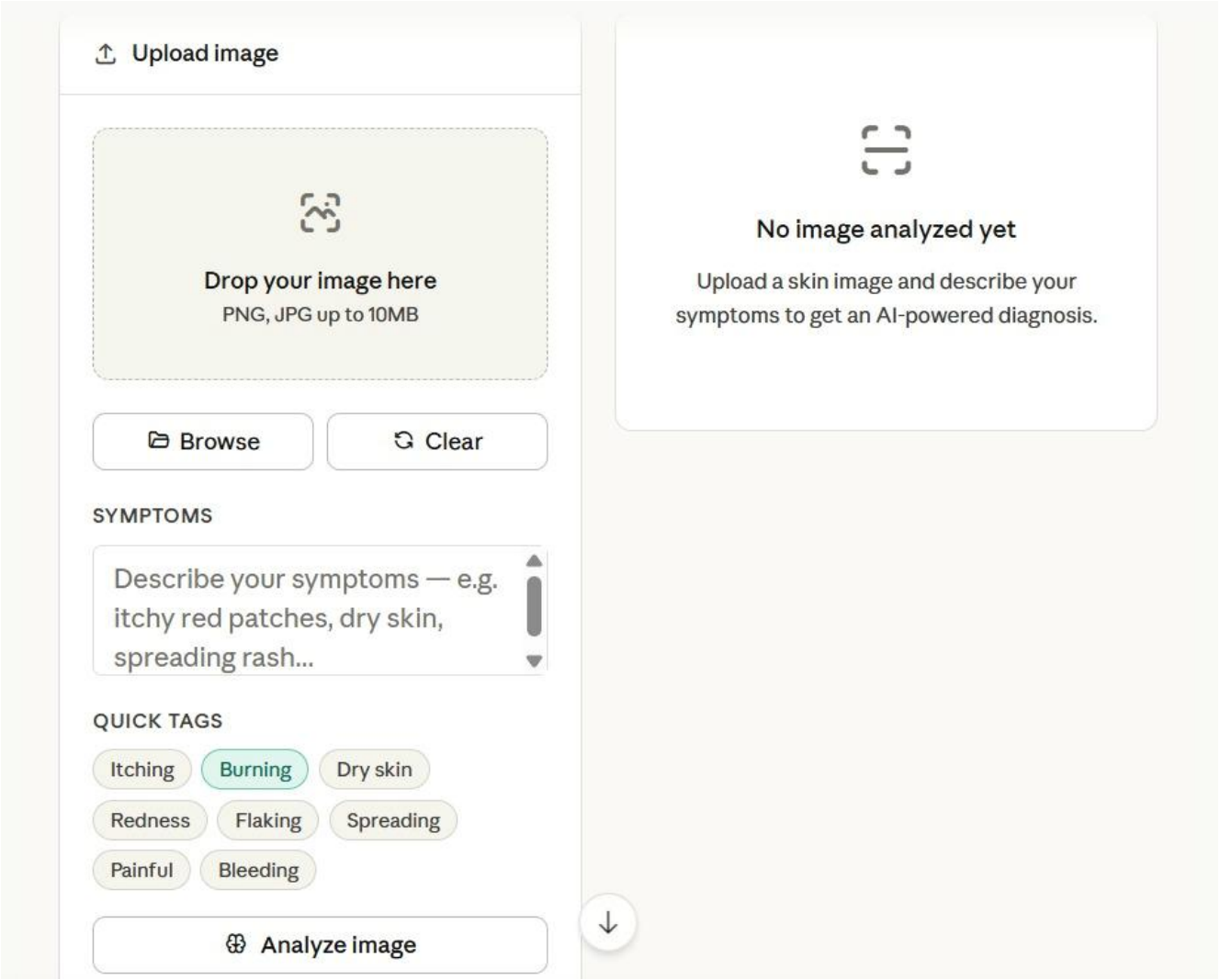


3.Activity Diagram

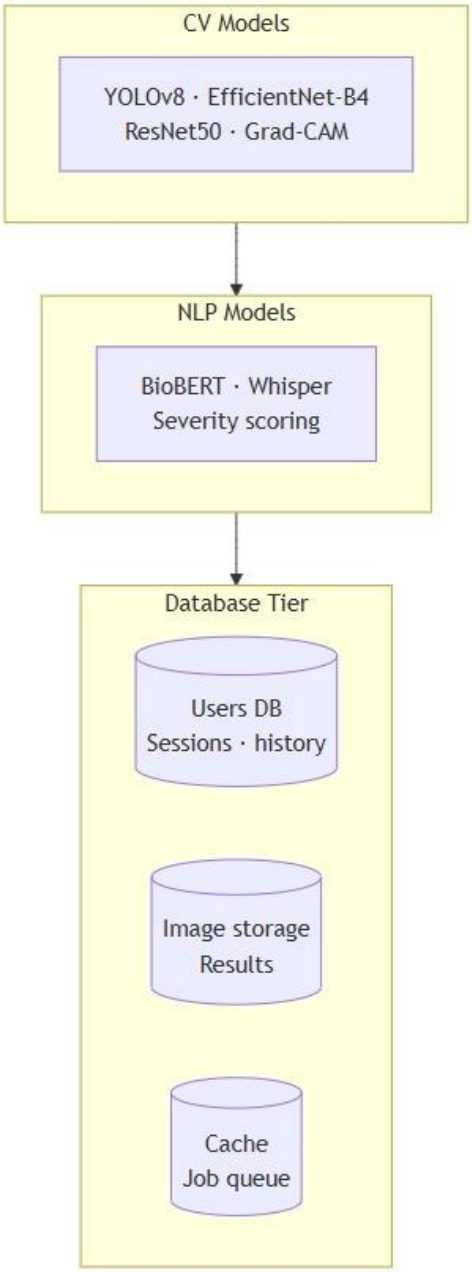


4.Class Diagram

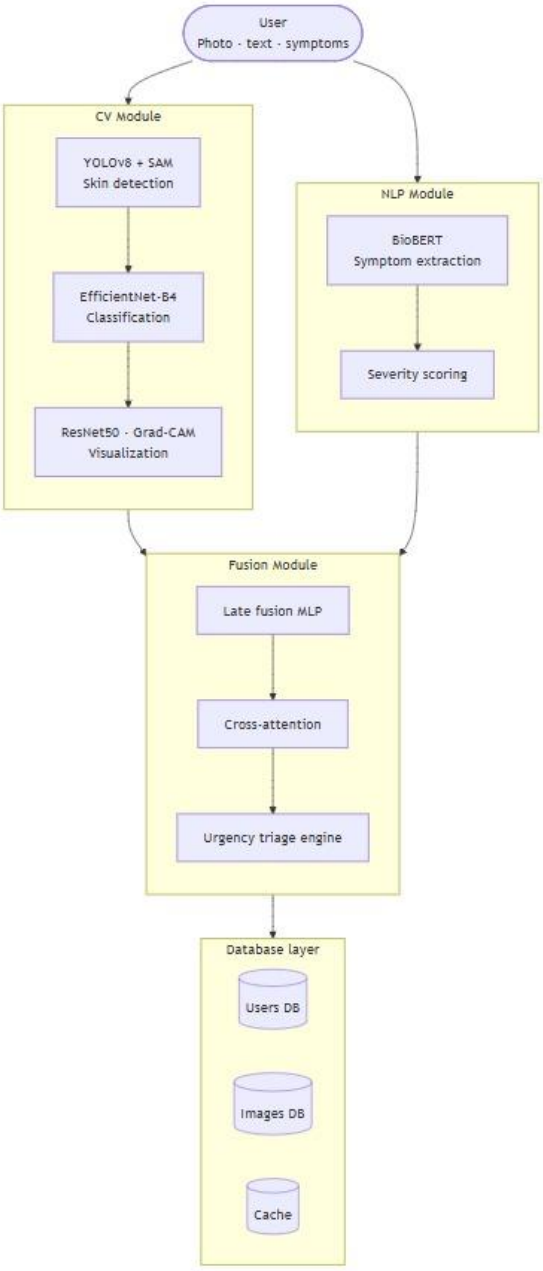
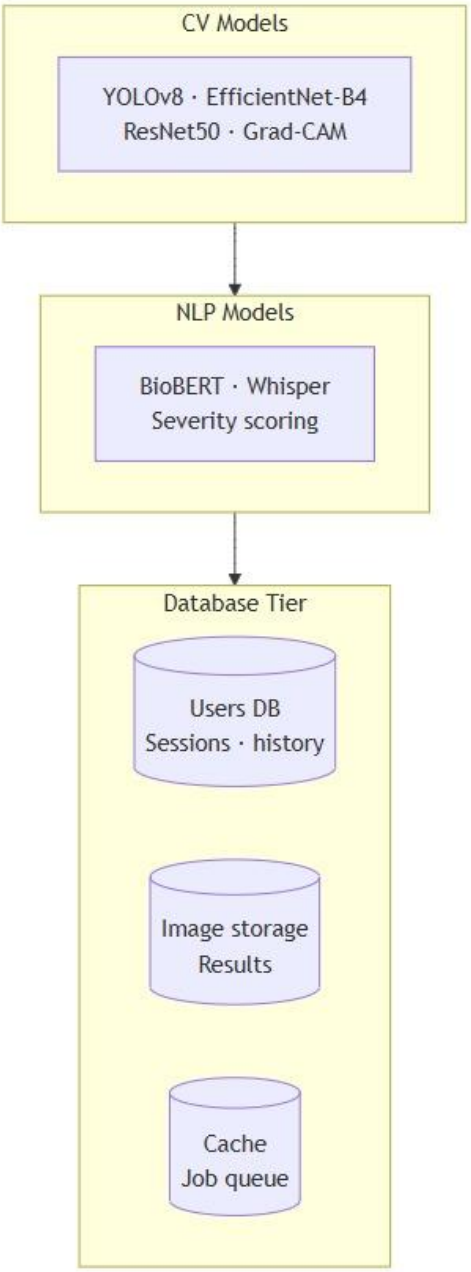



Upload image
Drop your image here
PNG, JPG up to 10MB
Browse
Clear
SYMPTOMS
Describe your symptoms — e.g. itchy red patches, dry skin, spreading rash...
QUICK TAGS
Itching
Burning
Dry skin
Redness
Flaking
Spreading
Painful
Bleeding
Analyze image
No image analyzed yet
Upload a skin image and describe your symptoms to get an AI-powered diagnosis.

1.Technology stack



2.Deployment diagram



3.Component Diagram